

ARJUN VARMA

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EDUCATION

Columbia University

Master of Science in Data Science

Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

New York, NY

Expected Dec 2026

Vellore Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering

Special Achiever Award | Merit Scholarship

Vellore, IN

May 2022

WORK EXPERIENCE

Novo Nordisk

Data Scientist Intern

Plainsboro, NJ

Jun 2026 - Present

- Identified and prioritized ~500 of 10,000+ U.S. HCPs for pre-launch planning for denecimig (Mim8), an investigational hemophilia therapy, using Komodo Healthcare Map real-world claims data to generate evidence on unmet need and market opportunity.
- Achieved 5.2x top-decile lift with an XGBoost propensity model trained on historical launches; translated model and rule-based signals into explainable HCP tiers and a dashboard used for nationwide field planning.
- Equipped field teams with tailored HCP engagement scripts and concise summaries of key HCP parameters by developing an LLM layer that synthesized access, efficacy, competitive-prescribing, and GLP-1 adoption signals with propensity scores and next best engagement model outputs.

ZS Associates

Advanced Data Science Associate Consultant

Pune, IN

Feb 2022 - Jun 2025

- Recovered missing categorical labels in medical transaction data with positive-unlabeled classification, expanding oncology customer-journey coverage from ~40% to 95% across tumor types and territories.
- Productionized the models with MLflow tracking, feature/prediction drift monitoring, and CI validation, reducing silent failures.
- Cut weekly reporting from days to minutes for 100+ stakeholders across \$10B in oncology brands by standardizing IQVIA and Symphony Health claims, Flatiron Health EHR-derived data, and Texas Oncology data in a Spark/SQL data warehouse with product and territory KPIs.
- Established audit-ready real-world evidence (RWE) analytics across \$4B+ in oncology products by engineering PySpark/SQL cohort pipelines over millions of healthcare records with logic resilient to missing and miscoded fields.
- Saved ~50 hours per month and raised first-pass quality above 99% by leading a five-member modernization of legacy analytics rules.

PROJECT EXPERIENCE

Biliary Tract Cancer Early-Detection Model - Python, PySpark, scikit-learn, NLP

- Accelerated identification by ~45 days relative to standard diagnosis timing by combining clinical risk factors, K-means/GMM segmentation, and transformer-based NLP clustering.
- Scaled early-detection modeling across 250M longitudinal claims by building PySpark cohorts and clinical risk-factor features that supported patient segmentation and model scoring.
- Drove territory-level resource planning with adopted model outputs; methodology from the project was presented at the PMSA 2025 conference.

DEUCE: Live Tennis Forecasting Platform - Python, XGBoost, Next.js, GitHub Actions

[GitHub](#) / [Live Demo](#)

- Powered live match forecasting and tournament simulation from 87,957 evaluated ATP/WTB matches by combining surface Elo, opponent-adjusted serve/return Markov features, and XGBoost.
- Improved leakage-free walk-forward Brier score from 0.2006 to 0.1950 (ATP) and from 0.2114 to 0.2017 (WTB); published frozen pre-match forecasts, model scorecards, and market comparisons.
- Operate live forecasts through hourly multi-source data refreshes, daily retraining, schema/freshness gates, failure alerts, Python/TypeScript CI, and automated Firebase deployment.

Filing Intelligence: Financial Research RAG - Python, FastAPI, ChromaDB, Next.js, GCP

[GitHub](#) / [Live Demo](#)

- Produced cited financial research across 15 companies and 4,967 passages from 128 filings, earnings decks, and call transcripts by building an evidence-first RAG workspace with scoped Chroma retrieval, deterministic ranking, and bounded context.
- Sustain reliable, auditable retrieval with page/line citations and 72 automated tests spanning retrieval, authentication, API contracts, and production builds; operate authenticated FastAPI/Next.js services, source-page viewers, and freshness checks.

ClaimReady: AI Small-Claims Packet Generator - Python, OpenAI Agents SDK, FastAPI, Next.js, GCP

[GitHub](#) / [Live Demo](#)

- Generated a court-ready small-claims packet from contracts, invoices, emails, and images in under 60 seconds through a four-agent workflow with structured handoffs and multimodal extraction.
- Operationalized an auditable workflow on Cloud Run by integrating legal RAG, live NY business-registry lookup, deterministic jurisdiction/damages tools, WebSocket traces, and ReportLab PDF generation into FastAPI/Next.js services.

SKILLS

Programming & ML: Python, SQL, R; pandas, NumPy, scikit-learn, PyTorch, XGBoost, SHAP, propensity scoring, probabilistic modeling

Applied AI: NLP, LLMs/RAG, LangChain, agentic workflows, multi-agent orchestration, LLM evaluation and guardrails, FastAPI, REST APIs

Data & MLOps: Spark/PySpark, Databricks, MLflow, production ML, ETL/ELT, warehousing, monitoring, Docker, CI/CD

Databases & Cloud: Postgres, Snowflake, DuckDB, ChromaDB; AWS (S3, EMR, Athena, SageMaker); GCP Cloud Run

Healthcare & RWE: RWD/RWE, longitudinal claims, EHR data, cohorts, patient journeys, HCP targeting, launch analytics